

Falsifiable Experimental Protocol

The Oxford Battery Hamiltonian — Half-Cell Validation

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1. Purpose

Validate the **geometric healing claim** of the Oxford Battery Hamiltonian: that a 300 Hz constitutional heartbeat + 4-phase prime-indexed pulse envelope applied to a Li-ion half-cell will suppress dendrite nucleation and flatten concentration gradients under conditions where classical Butler-Volmer kinetics predict runaway risk.

This protocol maps directly onto the 1-D continuum simulation boundaries (D_Li sweep, C-rate sweep) and provides falsifiability criteria.

2. Core Hypothesis

- H (Null):** The 300 Hz topological drive has no measurable effect on dendrite density, coulombic efficiency, or electrode spatial homogeneity compared to standard CCCV charging at identical C-rates.
- H (Helix):** The 300 Hz topological drive reduces dendrite density by 2 orders of magnitude, maintains coulombic efficiency >99.9%, and flattens concentration gradients ($\Delta c/c_{\text{max}} < 0.001$) under diffusion-limited conditions.

If H is not rejected, the geometric healing claim is falsified.

3. Cell Configuration

3.1 Half-Cell Architecture

- Working electrode:** Li-metal foil (99.9%, 50 μm thickness, Alfa Aesar)
- Counter electrode:** LiFePO (LFP) on Al current collector (areal capacity $\sim 2 \text{ mAh/cm}^2$)
- Electrolyte:** 1 M LiPF in EC/DMC (3:7 v/v) + 2 wt% VC additive
- Separator:** Celgard 2325 (25 μm polypropylene, porosity 55%)
- Cell format:** CR2032 coin cell or custom Swagelok T-cell (for in-situ optical access)

3.2 Topological Drive Hardware

- Signal generator:** Keysight 33500B or equivalent (arbitrary waveform, 300 Hz carrier)
- Envelope modulator:** Custom 4-phase prime-indexed pulse generator (ln 2 / ln 3 / ln 5 / Master Reset)

- **Power amplifier:** Linear amplifier (bandwidth >1 kHz, low distortion <0.1%) to superimpose AC on DC bias
- **Coupling:** Series capacitor + bias-T to inject 300 Hz AC onto the DC charge current without disturbing the potentiostat

3.3 Control Group

- Identical cell construction, identical C-rate, **standard CCCV profile** (no AC superposition)
- Same temperature, same electrolyte batch, same Li-metal source

4. Parameter Mapping (Simulation → Experiment)

Simulation Parameter	Experimental Analog	Target Value	Rationale
C-rate = 3C-5C	Charge current density	6-10 mA/cm²	Aggressive but commercially relevant
D_Li = 1.2×10 ⁻¹¹ m²/s	Room temperature (25°C)	T = 25°C	Baseline diffusion regime
D_Li = 5×10 ⁻¹¹ m²/s	Low temperature (-20°C)	T = -20°C	Cold-weather stress test
> 0.17	Local SOC < 83%	Monitor via operando Raman	Shear threshold for nucleation
R_max > 1.0	Dendrite nucleation	Post-mortem SEM + EIS	Proxy → measured morphology
Δc/c_max → 0	Concentration uniformity	Operando X-ray diffraction / neutron depth profiling	Spatial homogenization claim

5. Test Matrix

5.1 Primary Matrix (Falsification Core)

Run	Temperature	C-rate	Drive	N (cells)	Primary Metric
A	25°C	3C	Standard CCCV	5	Dendrite density (SEM)
B	25°C	3C	Helix 300 Hz	5	Dendrite density (SEM)
C	25°C	5C	Standard CCCV	5	Dendrite density (SEM)
D	25°C	5C	Helix 300 Hz	5	Dendrite density (SEM)

Table 2 – continued

Run	Temperature	C-rate	Drive	N (cells)	Primary Metric
E	-20°C	3C	Standard CCCV	5	Dendrite density + CE
F	-20°C	3C	Helix 300 Hz	5	Dendrite density + CE

5.2 Secondary Matrix (Mechanism Probing)

Run	Temperature	C-rate	Drive	Technique	Target
G	25°C	3C	Standard	Operando Raman	(x,t) mapping
H	25°C	3C	Helix	Operando Raman	(x,t) mapping
I	25°C	3C	Standard	EIS (1 MHz–10 mHz)	SEI resistance growth
J	25°C	3C	Helix	EIS (1 MHz–10 mHz)	SEI resistance growth

6. Measurement Protocols

6.1 Dendrite Quantification (Post-Mortem)

1. **Disassembly:** Disassemble cells in Ar-filled glovebox after 50 cycles.
2. **Washing:** Rinse Li-metal anode with DMC (3×) to remove residual electrolyte/salt.
3. **Drying:** Vacuum dry at 40°C for 12 h.
4. **Imaging:** SEM (Zeiss Sigma 300 or equivalent) at 5 kV, secondary electron detector.
5. **Quantification:**
 - Image 10 random fields of view per anode (500×500 μm each).
 - Segment dendrite structures via thresholding (ImageJ / Python skimage).
 - Report: **dendrite area fraction [%]**, **maximum dendrite length [μm]**, **number density [mm²]**.

6.2 Coulombic Efficiency

- Measure after every 10 cycles using galvanostatic cycling (Arbin or Bio-Logic).
- Report average CE over cycles 1–50.
- **Falsification threshold:** If Helix CE < 99.5% at any point, H is weakened.

6.3 Operando Concentration Mapping

- **Raman spectroscopy:** Map LiC / C peak ratio across electrode depth (confocal Raman, 532 nm laser, 1 μm spatial resolution).
- **Neutron depth profiling (NDP):** If available, direct Li concentration profile with ~10 nm depth resolution.
- **Target:** $\Delta c/c_{\text{max}} < 0.001$ in Helix arm vs. > 0.05 in Standard arm at 5C.

6.4 EIS — SEI Health

- Frequency sweep: 1 MHz → 10 mHz, 10 mV RMS perturbation.
- Fit to equivalent circuit: $R_{ohm} + (R_{SEI} || C_{SEI}) + Warburg + C_{dl}$.
- Track R_{SEI} growth over 50 cycles.
- **Prediction:** Standard R_{SEI} grows 2–5×; Helix R_{SEI} stable (geometric healing prevents SEI fracture/reformation).

7. Falsifiability Criteria

The geometric healing claim is **refuted** if ANY of the following are observed:

Criterion	Measurement	Threshold for Falsification
C1: No dendrite suppression	SEM area fraction	Helix dendrite density > 10% of Standard at same C-rate
C2: No CE improvement	Coulombic efficiency	Helix CE < 99.5% OR $\Delta(CE) < 0.5\%$ vs. Standard
C3: No gradient flattening	Operando Raman / NDP	Helix $\Delta c/c_{max} > 0.01$ at 5C
C4: SEI degradation	EIS R_{SEI}	Helix R_{SEI} growth > 2× baseline over 50 cycles
C5: Frequency insensitivity	Parameter sweep	No measurable difference between 300 Hz and 100 Hz / 1000 Hz drives

If **C1 OR C2 OR C3** is triggered, H is rejected. C4 and C5 are secondary (mechanism-specific) falsifications.

8. Power Calculation & Statistics

- **Effect size:** Simulation predicts ~5 orders of magnitude separation in risk proxy. Even with 2 orders of magnitude experimental attenuation, this is enormous.
- **Sample size:** N = 5 per group provides >99% power to detect a 2-order-of-magnitude difference (log-normal distribution, $\alpha = 0.05$, two-tailed).
- **Analysis:** Log-transform dendrite metrics (area fraction, length) before parametric testing (t-test or ANOVA). Non-parametric backup: Mann-Whitney U.

9. Safety & Controls

- **Thermal runaway:** Coin cells at 5C can exceed 60°C. Use thermal chamber with active cooling and IR monitoring.
- **Li-metal handling:** All disassembly in Ar glovebox (< 1 ppm H₂O, < 1 ppm O₂).
- **AC drive safety:** Ensure amplifier is galvanically isolated and current-limited to prevent short-circuit damage.

- **Blinding:** SEM imaging and dendrite quantification performed by operator blinded to drive condition (Standard vs. Helix).

10. Timeline

Phase	Duration	Deliverable
Cell fabrication	2 weeks	30 coin cells (5 per group + spares)
Baseline cycling	1 week	Confirm baseline CE and capacity
Primary matrix (A–F)	3 weeks	50-cycle data, post-mortem SEM
Secondary matrix (G–J)	2 weeks	Operando Raman, EIS
Data analysis	1 week	Statistical comparison, figure generation
Total	9 weeks	Go / No-Go decision on geometric healing

11. Expected Outcomes

If H is supported (Helix works):

- Dendrite area fraction: Helix < 0.1% vs. Standard > 10% at 5C
- Coulombic efficiency: Helix > 99.9% vs. Standard ~98% at 5C
- Concentration gradient: Helix $\Delta c/c_{\text{max}}$ < 0.001 vs. Standard > 0.05
- SEI resistance: Helix stable, Standard 3× growth over 50 cycles

If H is supported (Helix fails):

- No measurable difference in dendrite density, CE, or gradient between arms
- The 300 Hz drive is electrochemically inert or actively harmful (e.g., SEI oscillation)
- The Oxford Battery Hamiltonian requires fundamental revision or restriction to specific parameter regimes

12. References

1. Hope, S. *The Oxford Battery Hamiltonian: A Topological Derivation of Electrochemical Charge/Discharge via the Constitutional Master Reset*. Helix AI Innovations Inc., 2026.
2. Hope, S. *CURL curl: A Topological Shortcut to the Three-Body Problem and Constitutional AI Phase-Locking*. Zenodo, 2026. DOI:10.5281/zenodo.19459223.
3. Hope, S. *The Macroscopic Topological-Temporal Condensate*. Zenodo, 2026.
4. Bard, A. J. & Faulkner, L. R. *Electrochemical Methods: Fundamentals and Applications*. 2nd ed., Wiley, 2001.
5. Newman, J. & Thomas-Alyea, K. E. *Electrochemical Systems*. 3rd ed., Wiley, 2004.

GLORY TO THE LATTICE. THE SHAPE HOLDS.

End of Protocol.